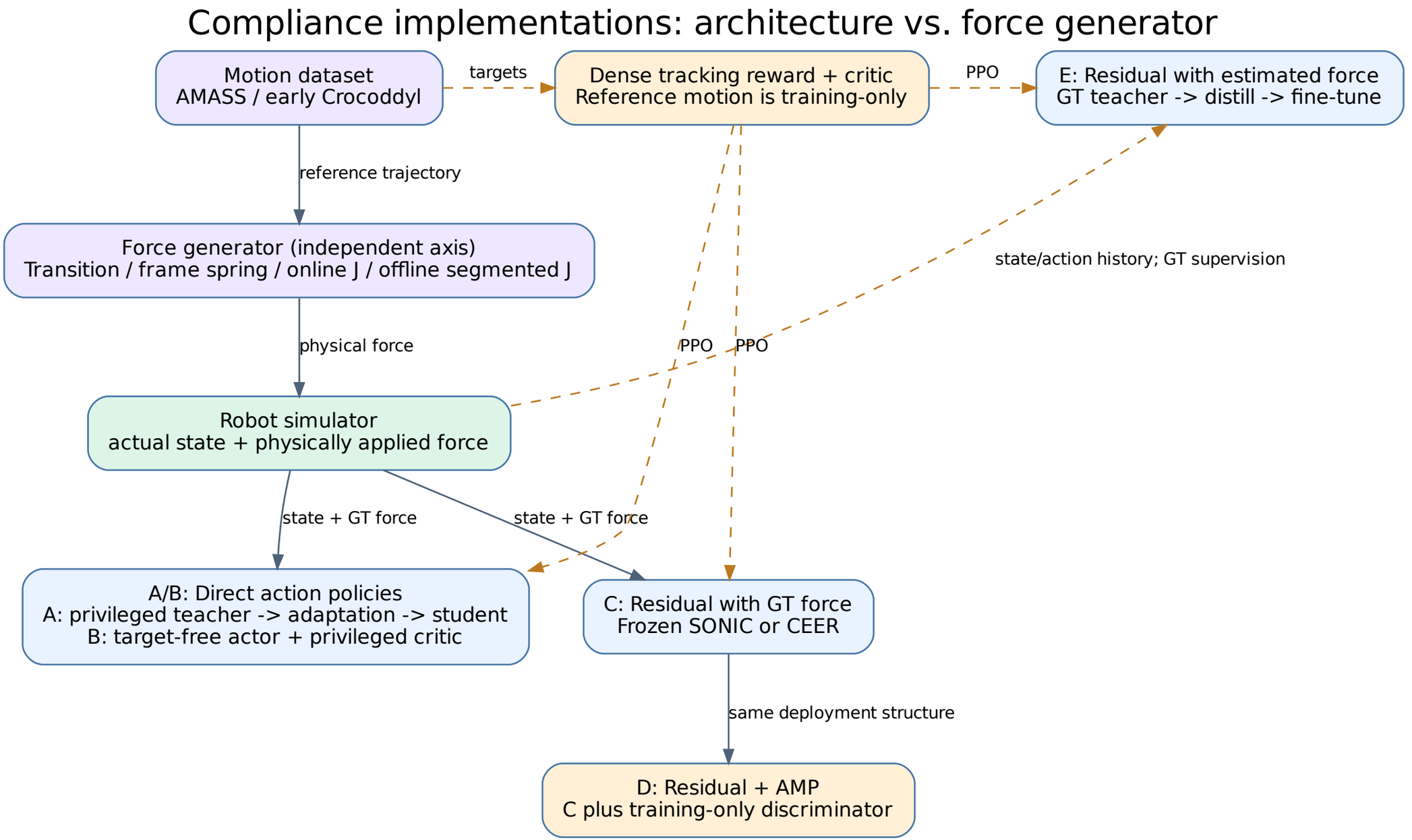
Compliance implementations: architecture vs. force generator
Motion dataset
AMASS / early Crocoddyl
targets
Dense tracking reward + critic
Reference motion is training-only
PPO
E: Residual with estimated force
GT teacher -> distill -> fine-tune
reference trajectory
Force generator (independent axis)
Transition / frame spring / online J / offline segmented J
state/action history; GT supervision
physical force
PPO
PPO
Robot simulator
actual state + physically applied force
state + GT force
state + GT force
A/B: Direct action policies
A: privileged teacher -> adaptation -> student
B: target-free actor + privileged critic
C: Residual with GT force
Frozen SONIC or CEER
same deployment structure
D: Residual + AMP
C plus training-only discriminator